

01



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SPACE SCIENCE & ASTRONOMY FOR OUTREACH & EDUCATION

Visualizing the Transient Sky: Real-Time All-Sky Imaging and Open Data in the TART Ecosystem

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02



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The TART Telescope



03



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CASSA Workshop 2 · Building & Learning TART



Students at the TART array



Working on TART's electronics



Prof. Tim Molteno lecturing



What the Next 13 Minutes Look Like

Interferometry: instead of one huge dish, link many small antennas — compare the waves each pair catches, and the maths turns them into a single giant telescope.

01

Feel the Interferometry

TART lets you build the radio sky by eye — no equations needed.

1. Two antennas, one interference pattern
2. One baseline makes one fringe
3. A fringe — the slice of sky a pair sees
4. How the fringe changes with distance and angle
5. A baseline becomes a point: the uv-plane
6. Calibration rotates the arrows
7. The conjugate twins, and the complete uv-map
8. Add the fringes → the sky appears

02

Run the Software Pipeline

The same picture, reproduced step by step from open data by open software.

1. Install the pipeline (Stimela)
2. Download raw visibilities
3. Create the measurement set
4. Label and safeguard the data
5. See what the array samples
6. Solve the amplitudes
7. Solve the phases
8. Correct the visibilities
9. Fourier-invert and CLEAN
10. The final sky

03

The Hardware

24 antennas, the correlator, the rooftop build.

Presented in the later talk



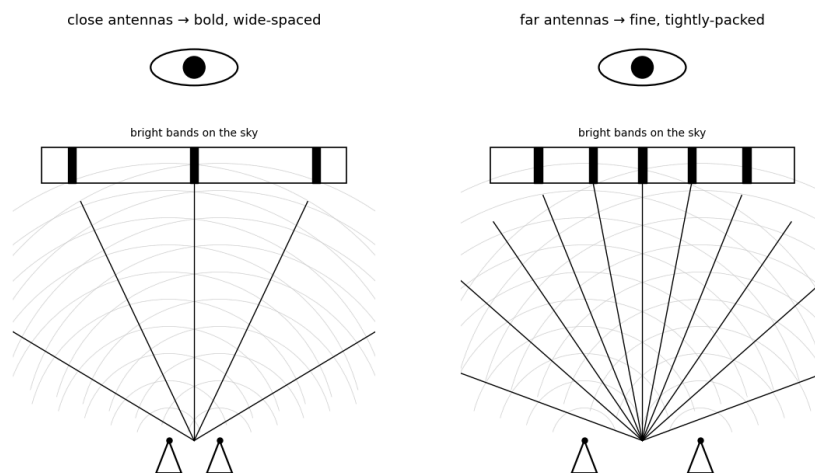
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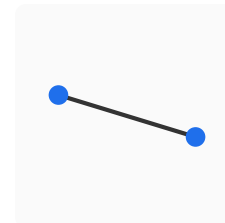
What a Fringe Is — and How It Changes with Distance and Angle

A pair of antennas never sees a picture. It sees one striped pattern laid across the whole sky. Move the pair further apart and the stripes get finer; turn the pair and the stripes turn with it — always at a right angle to the line joining the two antennas.



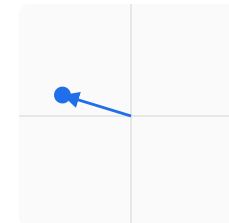
Wavefronts arriving at two antennas: where they add, a bright band; where they cancel, a dark one. Close pair → bold bands. Far pair → fine bands.

① the two antennas



1.20 m apart

② one dot in the uv-map



6.3 wavelengths from centre

③ one fringe on the sky



stripe spacing ≈ 0.16 (finer as the dot moves out)

Distance: **1.20 m**



Orientation: **163°**

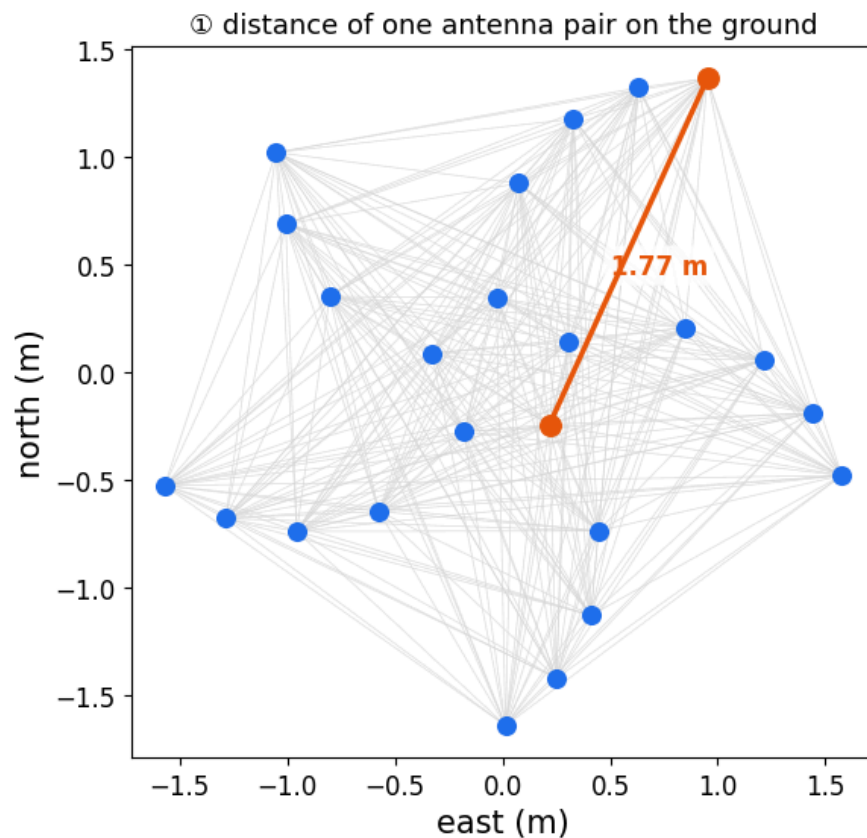


Slide **Separation** → the dot moves out and the stripes get finer. Slide **Orientation** → the dot circles the centre and the stripes turn with it, always at a right angle to the antenna pair.

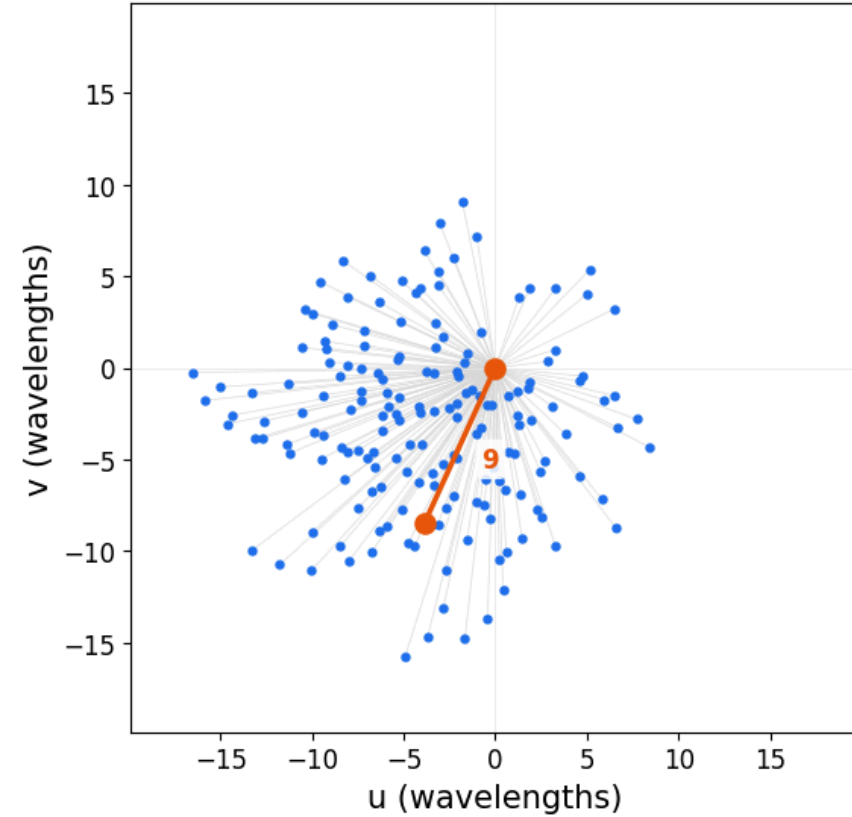
↑ These two sliders are live — scan the QR and drag them yourself.



A Baseline Becomes a Point



② one distance in m/wavelength $0.19\text{m} \rightarrow$ one baseline $\rightarrow$ a uv point

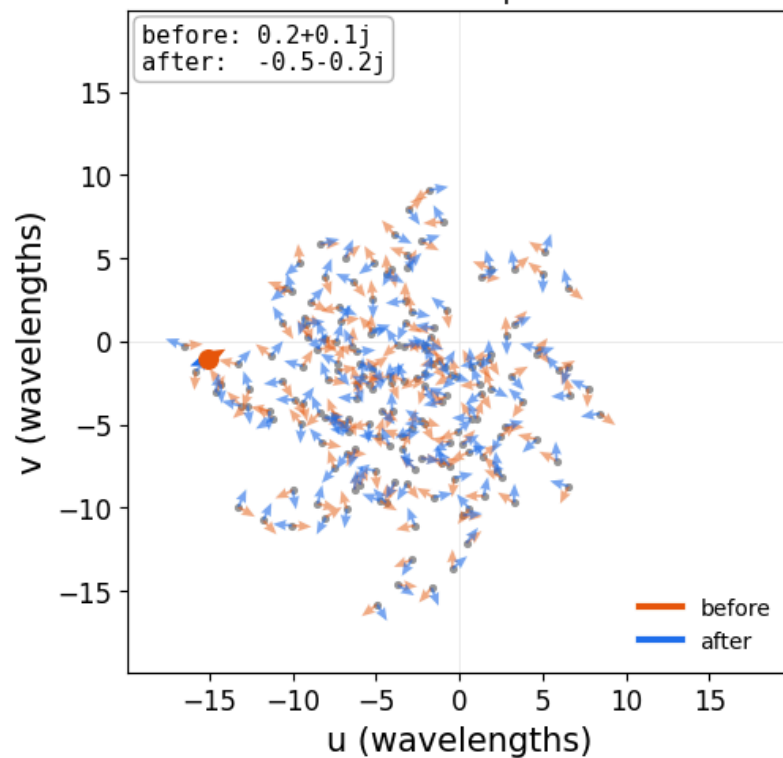




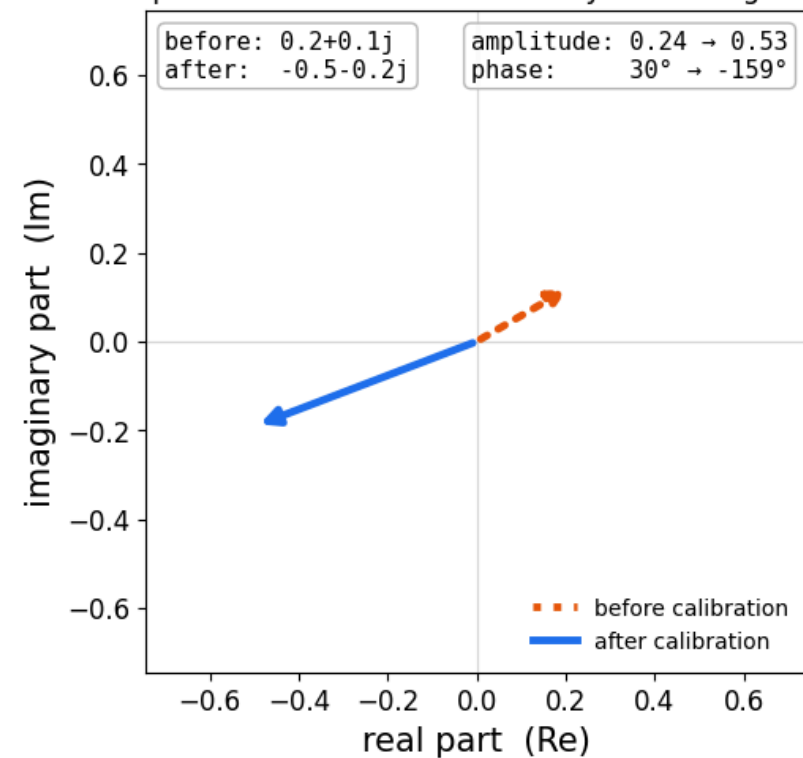
Calibration Rotates the Arrows

Each dot is a complex number (an arrow) — calibration changes its amplitude and phase

③ Calibration of complex visibilities



④ One point zoomed in — the visibility on the Argand plane

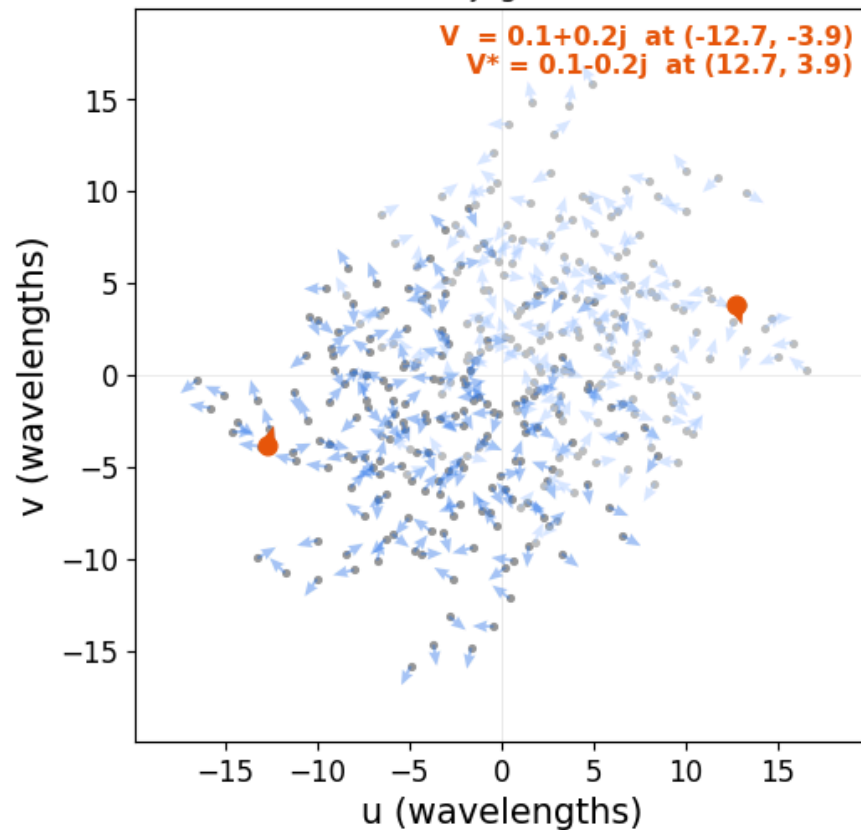




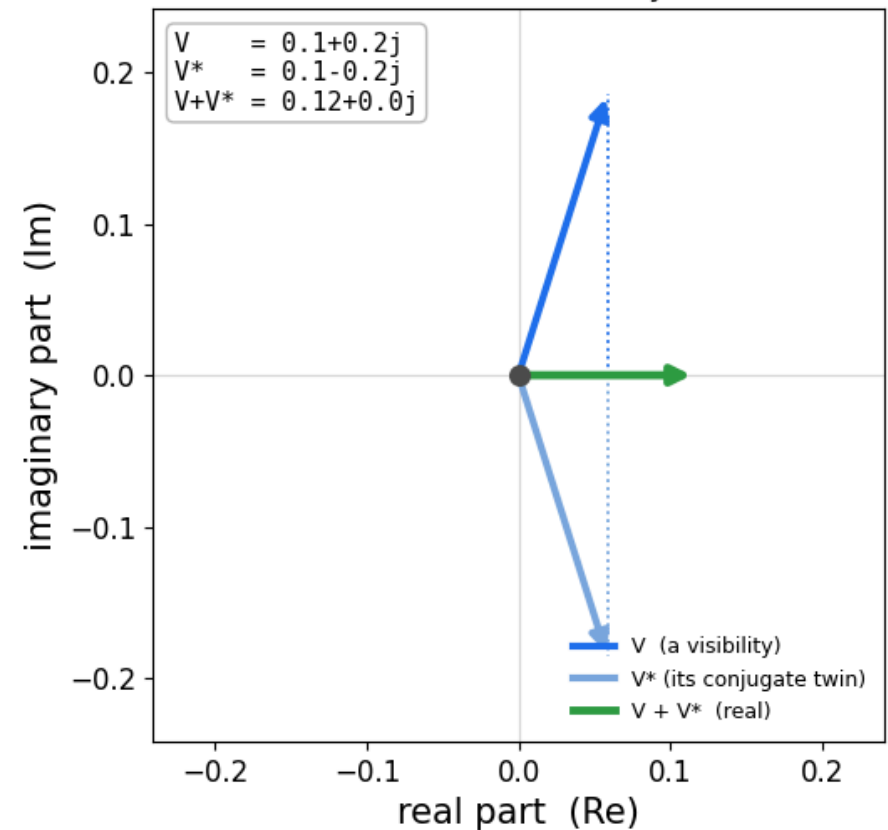
Corrected, Plus the Conjugate Twins

The conjugate twin cancels the imaginary part — leaving a real sky

⑤ + conjugate twins



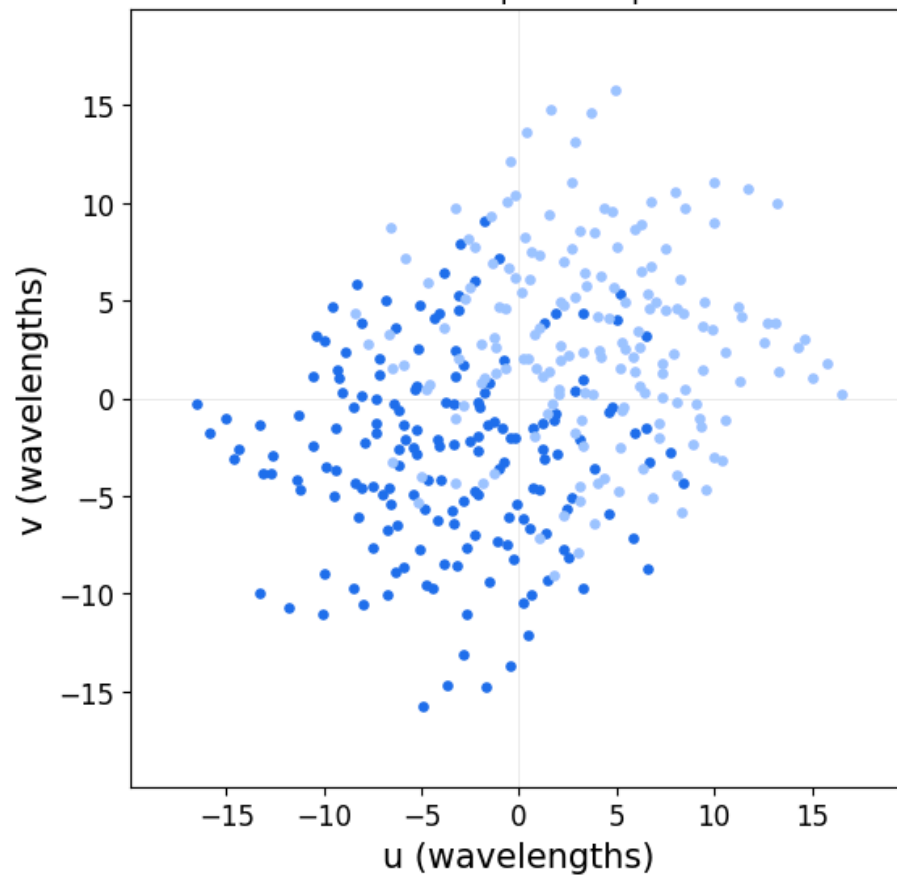
⑥ $V + V^*$ is real — the sky is real



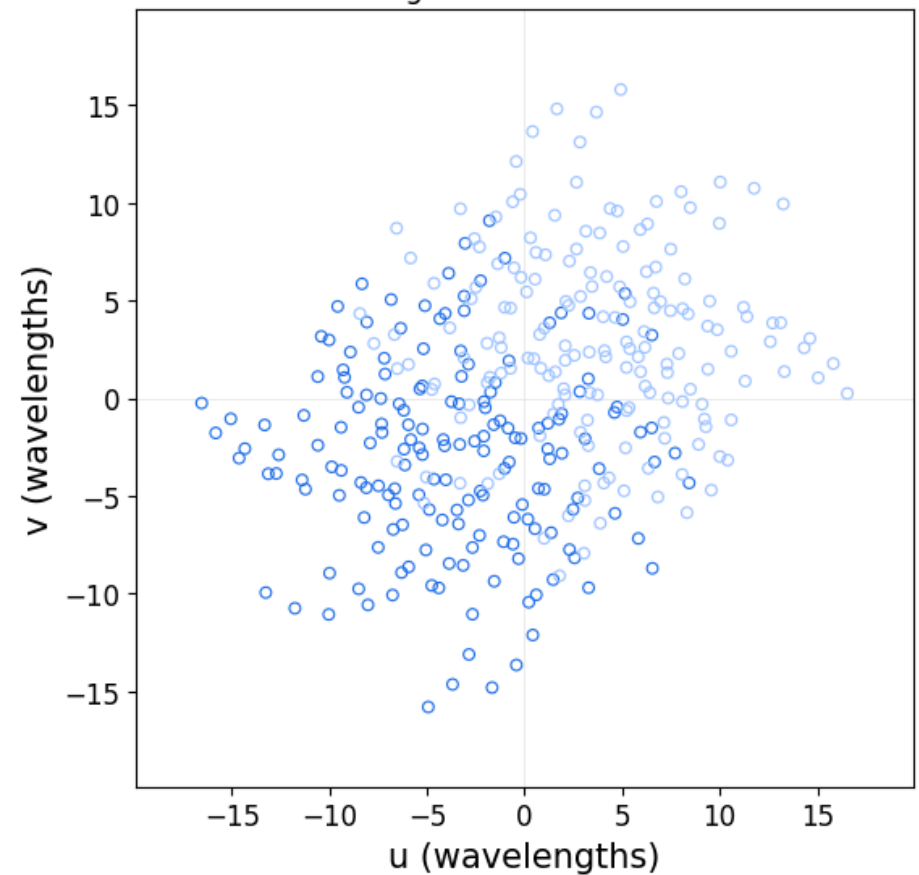


The Complete uv-Map — and the To-Do List

⑦ the uv-map — 342 points



⑧ 342 fringes to add — the to-do list





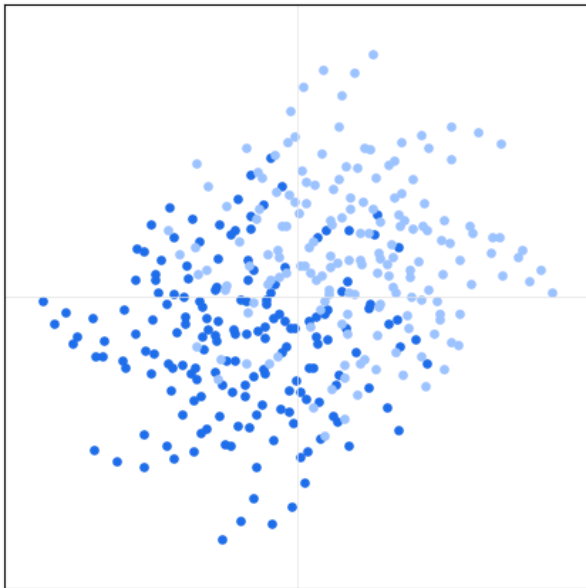
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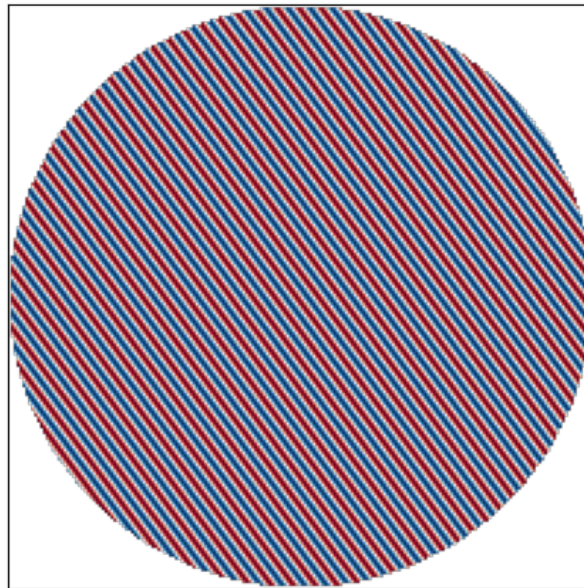
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Add the Fringes → the Sky Appears

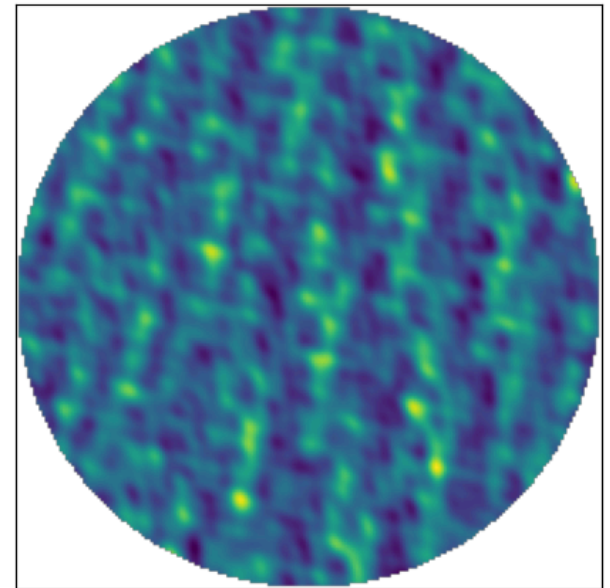
171/171 baselines used



the fringe just added



all fringes added → the sky



drag → add baselines one by one



SETUP

Install the Necessary packages and recipes

WHY

- Stimela runs each step inside a container.
- Reproducible on any WSL-2 machine.
- Install it once, then just run recipes.
- Every file and the full instructions live in one repo — clone it and follow along.

START HERE

```
git clone https://github.com/gwynbleiddrivia/tart-stimela-run
```

REQUIREMENTS

```
WSL 2
Apptainer 1.4.4 (+ suid)
squashfuse, fuse2fs, gocryptfs
Python venv: tart_cargo, cult_cargo, stimela
Recipe files: tart_dl.yaml, casacabs.yaml, casa/
```

INSTALL

```
sudo apt install -y ./apptainer_1.4.4_amd64.deb
sudo dpkg -i ./apptainer-suid_1.4.4_amd64.deb
sudo apt install -y squashfuse fuse2fs gocryptfs
python3 -m venv start && source start/bin/activate
pip install tart_cargo cult_cargo stimela
cp
/mnt/d/tart/{tart_dl.yaml,casacabs.yaml,make_mov.py}
.
cp /mnt/d/tart/make_mov.py .
cp -r /mnt/d/tart/casa/ ./casa/
```



STAGE 1 • ACQUIRE

Download Raw Visibilities

WHAT THIS DOES

- Interferometry starts from correlations, not images.
- Each antenna pair → one complex visibility.
- Pull a batch of snapshots off the telescope.
- <https://api.elec.ac.nz/tart/bd-iub/api/v1/info>
- <https://api.elec.ac.nz/tart/bd-iub/api/v1/mode/current>
- <https://api.elec.ac.nz/tart/bd-iub/api/v1/calibration/gain>
- <https://api.elec.ac.nz/tart/bd-iub/api/v1/imaging/vis>

ALGORITHM

```
make the working folders
ask the TART API for N raw snapshots
each snapshot = 276 baselines × complex vis
```

COMMAND

```
mkdir stimela_images img rawdata caltables msdir
stimela run tart_dl.yaml tart=bd-iub -s download-hdf
```




STAGE 2 • BUILD

Create the Measurement Set

WHAT THIS DOES

- Raw JSON → the standard radio dataset (MS).
- Compute each baseline's (u,v) geometry.
- Predict the known-satellite sky model.

ALGORITHM

```
baseline (u,v) = position_j - position_i
fetch GNSS catalogue (known positions)
predict their model visibilities
rephase everything to the zenith
```

COMMAND

```
stimela run tart_d1.yaml tart=bd-iub -s create-ms
```



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STAGE 3 • PREP

Label & Safeguard

WHAT THIS DOES

- Tag the data so standard tools accept it.
- Save a restore-point before we alter anything.
- Every later step stays reversible.

ALGORITHM

```
rename observatory → CASA/WSClean accept it  
snapshot the current flags as 'ORIGINAL'
```

COMMAND

```
stimela run tart_dl.yaml tart=bd-iub -s  
updateobservatory  
stimela run tart_dl.yaml tart=bd-iub -s flagsave
```



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STAGE 4 • INSPECT

See What the Array Samples

WHAT THIS DOES

- The baselines ARE the Fourier sampling.
- Plot the uv-coverage and antenna layout.
- This sparse set of points is all we know.

ALGORITHM

```
plot each baseline as a point in the uv-plane
plot the 24 antenna positions
print an observation summary
```

COMMAND

```
stimela run tart_dl.yaml tart=bd-iub -s plotuv
stimela run tart_dl.yaml tart=bd-iub -s plotants
stimela run tart_dl.yaml tart=bd-iub -s lister
```



STAGE 5 • CALIBRATE

Solve the Amplitudes

WHAT THIS DOES

- Each antenna has an unknown gain.
- Match the data to the known-satellite model.
- Normalise mean gain to 1 (no absolute flux).

ALGORITHM

```
find gain a_p so |a_p·a_q|·MODEL ≈ DATA  
average over the snapshot  
normalise mean |gain| → 1
```

COMMAND

```
stimela run tart_dl.yaml tart=bd-iub -s  
calibrate_amplitude  
stimela run tart_dl.yaml tart=bd-iub -s  
plotcaltable_amp
```



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STAGE 6 • CALIBRATE

Solve the Phases

WHAT THIS DOES

- Each antenna's clock is off by an unknown phase.
- That's why sources don't focus.
- Solve the offsets so fringe crests line up.

ALGORITHM

```
baseline phase = true phase + (clock_p - clock_q)
solve each antenna's clock offset (every 10 s)
subtract it
```

COMMAND

```
stimela run tart_dl.yaml tart=bd-iub -s
calibrate_phase
stimela run tart_dl.yaml tart=bd-iub -s
plotcaltable_phase
```




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STAGE 7 • APPLY

Correct the Visibilities

WHAT THIS DOES

- Divide the solved gains out of every visibility.
- Produces CORRECTED data, ready to image.
- Dead antennas dropped — never ÷ by zero.

ALGORITHM

```
CORRECTED = DATA / (gain_p · conj(gain_q))  
drop flagged / dead-antenna baselines
```

COMMAND

```
stimela run tart_d1.yaml tart=bd-iub -s applycal
```



STAGE 8 • IMAGE

Fourier-Invert & CLEAN

WHAT THIS DOES

- Fourier-sum the visibilities → the 'dirty' image: the true sky smeared by the array's messy beam.
- Gaps in the uv-map make every source drag a splatter of rings and spikes (sidelobes).
- CLEAN finds the brightest spot, records it as a point source, and subtracts that spot's known splatter — over and over.
- Finally it repaints the recorded sources with one smooth, clean beam → the final image.

HOW CLEAN WORKS

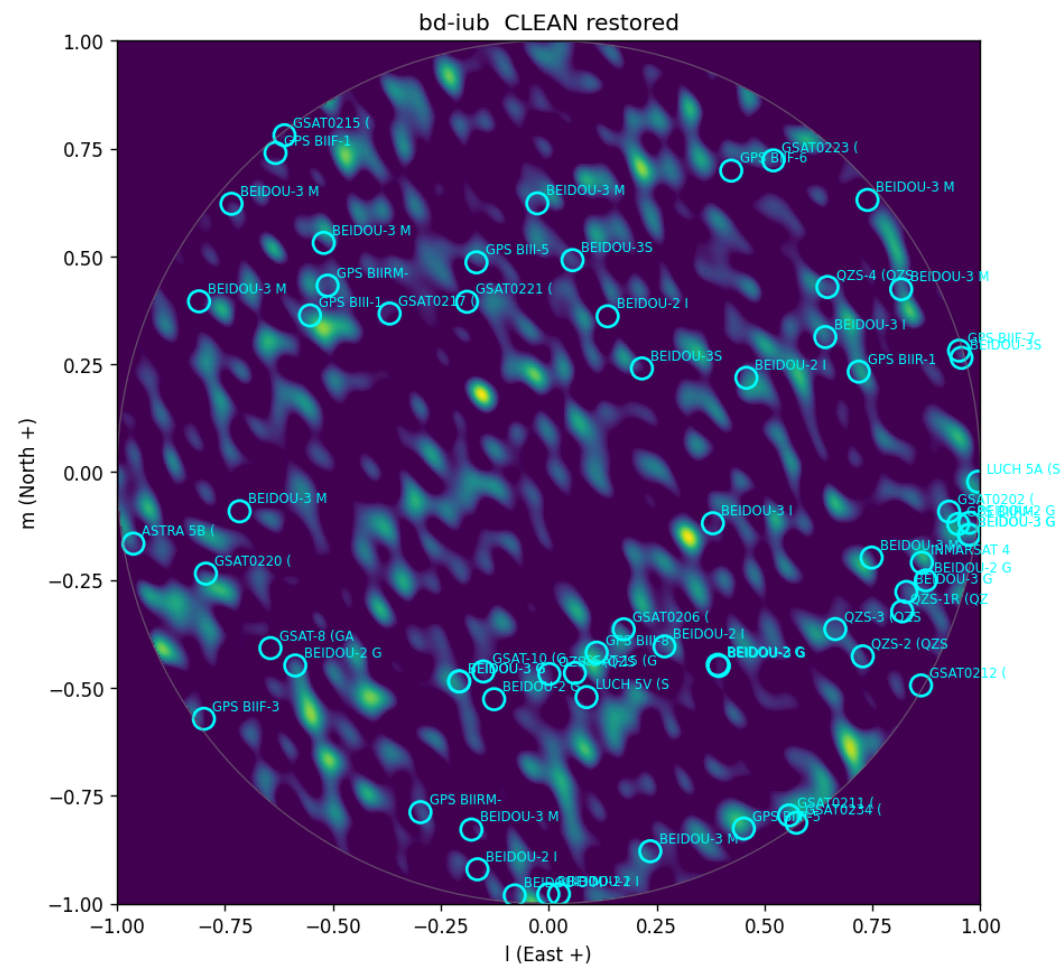
```
dirty image = true sky * dirty beam (PSF)
loop:
  find the brightest pixel
  add a fraction of it to the model
  subtract that fraction × shifted PSF
until only noise remains
restore: model * clean beam + leftover noise
```

COMMAND

```
stimela run tart_dl.yaml tart=bd-iub -s
snapshotimage
```

- or visit -

The Final Sky



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Under One Sky

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